

Clinical utility of loneliness as a symptom of major depression in youth

João Villanova do Amaral^{1,2}, Gabriele dos Santos Jobim^{2,3}, João Pedro Gonçalves Pacheco^{1,2,3}, Lucas Toshio Ito^{4,14}, Rafaella Ormond^{4,14}, Marcos Leite Santoro^{4,7,14}, SÍntia Iole Belangero^{4,5,7}, Ary Gadelha^{5,6,7}, Euripedes Constantino Miguel^{6,7,8}, Rodrigo Affonseca Bressan^{4,5,6,7}, Pedro Mario Pan^{4,5,6,7}, Luis Augusto Rohde^{1,6,7,9,10,11}, Giovanni Abrahão Salum^{1,6,7,9,12}, Maurício Scopel Hoffmann^{1,2,3,7,13}

¹ Graduate Program in Psychiatry and Behavioral Sciences, Universidade Federal do Rio Grande do Sul, Rua Ramiro Barcelos 2350, Porto Alegre, 90035-003, Brazil (UFRGS), phone +55-51-3308-5624

² Mental Health Epidemiology Group (MHEG), Universidade Federal de Santa Maria, Santa Maria, RS, Brazil

³ Department of Neuropsychiatry, Universidade Federal de Santa Maria, Avenida Roraima 1000, building 26, office 1353, Santa Maria, 97105-900, Brazil (UFSM), phone +55-55-3220-8574

⁴ Laboratório Neurociências Clínicas Integrativas, Departamento de Psiquiatria, Universidade Federal de São Paulo (UNIFESP), São Paulo, SP, Brazil

⁵ Department of Psychiatry, Escola Paulista de Medicina, Universidade Federal de São Paulo, R. Maj. Maragliano, 241 - Vila Mariana, São Paulo - SP, 04017-030 Brazil (UNIFESP)

⁶ National Institute of Developmental Psychiatry for Children and Adolescents (INCT-CNPq), São Paulo, Brazil

⁷ National Center for Innovation and Research in Mental Health (CISM), São Paulo, Brazil

⁸ Department of Psychiatry, Faculdade de Medicina, Universidade de São Paulo, Rua Dr. Ovídio Pires de Campos 785, São Paulo, 01060-970, Brazil (USP)

⁹ Department of Psychiatry and Legal Medicine, Universidade Federal do Rio Grande do Sul, Rua Ramiro Barcelos 2350, Porto Alegre, 90035-003, Brazil

¹⁰ ADHD Outpatient Program & Developmental Psychiatry Program, Hospital de Clínicas de Porto Alegre, Federal University of Rio Grande do Sul, Brazil

¹¹ Medical Council UNIFAJ & UNIMAX, Brazil

¹² Child Mind Institute, New York, NY 10022, USA

¹³ Care Policy and Evaluation Centre, London School of Economics and Political Science, Houghton St, London WC2A 2AZ, UK.

¹⁴ Department of Biochemistry, Escola Paulista de Medicina, Universidade Federal de São Paulo, R. Tres de Maio, 100, - Vila Clementino, São Paulo - SP, 04044-020 Brazil (UNIFESP)

Corresponding author: Dr. Mauricio Scopel Hoffmann. Universidade Federal de Santa Maria, Avenida Roraima 1000, building 26, office 1353, Santa Maria, 97105-900, Brazil (UFSM), phone +55-55-3220-8574; e-mail: mauricio.hoffmann@ufsm.br. I declare that I have full access to all the data in the study and had final responsibility for the decision to submit it for publication

Abstract

Background: This study aims to examine the utility of adding loneliness as a symptom of major depressive disorder (MDD).

Methods: We used data from the 2018-2019 wave of the Brazilian High-Risk Cohort Study, involving 1,905 participants aged 14-23. We evaluated the centrality of loneliness within the depressive symptom network. Finally, we used adjusted multiple regression (sociodemographic factors and social isolation) to examine the association of 1) loneliness alongside classical symptoms of MDD and 2) clinical groups (including or excluding loneliness), with a composite of life-threatening events (suicide attempt, self-harm, psychiatric admission, or emergency room visits), poor life-chance events (criminal convictions, substance use, and school dropout), and well-being.

Results: Loneliness was part of the depressive construct and the most central symptom in the network structure of depressive symptoms. It was independently associated with life-threatening events (OR: 1.754; 95% CI: 1.187, 2.590), life chance impoverishment events (OR: 1.446; 95% CI: 1.065, 1.963), and well-being (B: -2.681; 95% CI: -3.792, -1.569). Lastly, the group with MDD that included loneliness as a symptom had a higher risk for life-threatening events (OR: 3.168; 95% CI: 2.039, 4.976) and had lower scores on well-being (B: -3.474; 95% CI: -5.085, -1.862), compared to those with MDD without loneliness.

Conclusion: Loneliness is a constitutive symptom of depression that serves as an indicator of severity in MDD. The operationalization of loneliness as a clinical criterion of MDD is expected to improve clinical care of young people with depression.

Introduction

Major depressive disorder (MDD) is a clinical syndrome marked by dysfunctions across distinct psychobiological dimensions, causing functional impairment (Zachar and Kendler, 2017). Different models explain that MDD symptoms stem from a common underlying factor or may reinforce each other (Aristodemou et al., 2024). It is explained by multiple risk factors, none of which is enough to account for its etiology (Thapar et al., 2022). In the past years, studies have shown that MDD is closely linked with loneliness, especially in adolescents and young adults (Achterbergh et al., 2020; Cacioppo et al., 2006; Erzen and Çikrikci, 2018; Mann et al., 2022; Wang et al., 2018). In fact, both quantitative and qualitative studies conclude that loneliness is central to the experience of being depressed (Fusar-Poli et al., 2023; Manfro et al., 2021; Viduani et al., 2024, 2021). For instance, loneliness and social withdrawal are recurrent features in young individuals with lived experience of MDD, who feel disconnected from the world. As stated by an adolescent: “I just feel lonely, like nobody understands what I’m going through” (Midgley et al., 2015; Viduani et al., 2024). However, loneliness is not considered as a clinical indicator of this mental health condition.

Loneliness is a major concern in public health (Holt-Lunstad, 2022; Lim et al., 2020; Luhmann et al., 2023; Solmi et al., 2020). Defined as the subjective feeling of being alone and without social support, loneliness should be distinguished from social isolation, a quantitative phenomenon characterized by living alone and having few social relationships, among other aspects (Perlman and Peplau, 1981). A recent meta-analysis shows that the prevalence of loneliness is higher in adolescents (6.8% to 17.1%) and in the elderly (18.7% to 24.2%), despite the significant variation among the analyzed countries (Surkalim et al., 2022). Aside from its high prevalence, loneliness has an impact on physical and mental health (Beutel et al., 2017; Hodgson et al., 2020; Holt-Lunstad and Perissinotto, 2023; Leigh-Hunt et al., 2017; Luo et al., 2012; Valtorta et al., 2016; Wang et al., 2023), as well as on the socioeconomic future of young

adults (Bryan et al., 2024), being a risk factor for early mortality (Holt-Lunstad et al., 2015). Despite being a distressful feeling, loneliness receives little attention in the Diagnostic and Statistical Manual of Mental Disorders (DSM) and in the International Classification of Diseases (ICD-11) (World Health Organization). Considering its latest published edition, loneliness is only cited as a symptom associated with depression across culture-related contexts (American Psychiatric Association, 2022).

From a molecular perspective, evidence demonstrates that depression and loneliness are intimately close, which might explain their longitudinal bidirectional association (Abdellaoui et al., 2018; Abdellaoui and Verweij, 2021; Choi et al., 2020; Sbarra et al., 2023). In addition to the molecular level, loneliness is also clinically associated with depression (Manfro et al., 2021; Mann et al., 2022; Sbarra et al., 2023; Wang et al., 2018). Recent evidence using a network analysis of depressive symptoms demonstrates that loneliness is a central symptom in adolescents, even among other MDD diagnostic criteria (Manfro et al., 2021; Mullarkey et al., 2019). In fact, interventions targeting social support have a protective effect on depression (Gariépy et al., 2016). This demonstrates the important role of feelings regarding social connection in depressive symptomatology, especially when considering young people. Despite that, it remains unclear whether loneliness is a good indicator of depression.

Therefore, the aim of the current study is to evaluate whether loneliness is part of the latent construct of depression, replicate its centrality within the depressive syndrome, and advance the field by investigating whether it serves as a predictive symptom and severity indicator when operationalized alongside other DSM criteria. Based on previous findings, we hypothesize that 1) loneliness will be part of the depression latent construct and will be central in a network of depressive symptoms, 2) loneliness will be a useful symptom while having predictive validity above and beyond other depressive symptoms and 3) loneliness will be a severity indicator in depression.

Methods

Study design and participants

We analyzed data extracted from the second wave (2018-2019) of the Brazilian High-Risk Cohort Study (BHRCS). The BHRCS is a school-based cohort oversampled for high family risk for psychopathology (Salum et al., 2015). Families were recruited from 22 state-funded schools in Porto Alegre and 35 schools in São Paulo on the day of enrolment, between January and March 2010. The second wave included 1.905 participants (76.3% retention, 14-23 years of age, 46.9% females). All parent-reports were collected by lay interviewers and self-reported outcome measures were collected by trained psychologists (see below for specific measures).

This study was approved by the National Research Ethics Commission (Comissão Nacional de Ética em Pesquisa) under the approval number 2.448.062 (CAAE 74563817.7.1001.5327), University of São Paulo and Federal University of Rio Grande do Sul ethics committees. Written informed consent was obtained from parents and participants. From those who were illiterate, the consent was read, doubts were explained, and verbal agreement was obtained. None were excluded due to parental illiteracy (Salum et al., 2015).

The analysis was structured in four steps: First, using confirmatory factor analysis (CFA), we evaluated if loneliness aggregates in depression and established loneliness' clinical threshold (see below). Second, using the same questionnaire, we examined the centrality of loneliness among other MDD symptoms (see network analysis below). Third, to understand the clinical utility of loneliness compared to other depressive symptoms present in the DSM - decoded from DAWBA, the clinical validators (poor life chances, life-threatening and well-being) (see below) by each symptom (loneliness validity among other depression symptoms). Fourth, we regressed these clinical validators by different clinical groups (individuals without

depression nor loneliness, with loneliness without MDD and with MDD including loneliness) compared with those with depression without loneliness, to evaluate if the presence of loneliness as a depressive symptom would indicate higher severity (loneliness as a severity indicator).

Psychiatric assessment

Depressive symptoms and Major depressive disorder

Depressive symptoms were assessed using the Brazilian-Portuguese self-report version (Pinto, 2014) of the Short Mood and Feelings Questionnaire (SMFQ) and the Development and Well-Being Assessment (DAWBA) (Fleitlich-Bilyk and Goodman, 2004; Goodman et al., 2000). The SMFQ was developed as an easy-to-administer self-reported tool for measuring depression. It asks about 13 symptoms in the past two weeks, including loneliness, using a 3-point likert scale (0 = Not true; 1 = Sometimes; 2 = True) (Angold, Costello, Messer, & Pickles, 1995).

Major depressive disorder was assessed with DAWBA, a semi-structured interview used to generate diagnostic probabilities in accordance with DSM-5 and ICD-10. It is composed of skip rules (e.g., participants will only be asked about all MDD symptoms if considered positive for depressed/irritable mood or loss of interest) and open-ended questions. Interviews were conducted by trained psychologists directly with the participants, with different questionnaires applied to adolescents and young adults. Based on participant's responses, DAWBA provides a computer-generated category that informs the probability of a positive diagnosis. This probability and open-ended responses were evaluated by trained psychiatrists who confirmed, refuted, or altered initial DAWBA bands to determine the final diagnosis. DAWBA was decoded in 14 items for young adults and 17 items for adolescents, with the additional items in the latter group attributed to the irritability section. These items

were combined to mirror the nine DSM-based MDD symptoms (Table S1). DSM-based symptoms were used to analyze the predictive validity of each symptom, including loneliness (see below).

Loneliness

Loneliness was derived from the SMFQ (item 10 - “I felt lonely”). We first estimated a SMFQ factor model (see confirmatory factor analysis section below). Then, we binarized the item “I felt lonely” using the threshold of 0.5. This value represents a half standard deviation above the mean of the distribution of subjects with feeling lonely over the past two weeks. Therefore, we interpreted values < 0.5 as typical development (normative) and values ≥ 0.5 as clinical indicators, compatible with DAWBA-based depression items, which are also binary. For example, if the threshold ≥ 0.5 is the threshold between the scores 0 (“Never”) and 1 (“Sometimes”), we consider those with scores ≥ 1 as positive for loneliness.

Clinical Groups

To evaluate the impact of including loneliness as a MDD symptom, we used the DAWBA-based clinician diagnosis and the presence of loneliness as a symptom to establish four clinical groups. They were composed of individuals (1) with major depression and loneliness as a component of the syndrome; (2) with depression, but without loneliness; (3) without depression, but lonely; (4) neither lonely nor depressed.

Clinical Validators

Life-threatening validator

Life-threatening validator is a composite of any positive cases of either (1) self-report suicide attempt in the past four weeks or self-harm requiring medical attention in the past six

months, (2) parent-reported psychiatric inpatient admission or (3) emergency visits due to emotional or behavioral problems.

Data on self-harm was gathered using the Deliberate Self-Harm Inventory (DSHI), which inquired: "In the last six months, have you engaged in any self-injurious behavior that necessitated hospitalization or medical treatment?". The responses ranged from "once" to "more than five times".

The items used to evaluate parent-reported psychiatric hospitalizations and emergency visits due to emotional or behavioral issues were as follows: "Since the last assessment, was your child admitted in a hospital or clinic for emotional or behavioral problems?" and "Since the last assessment, has the child ever received outpatient help from an emergency room for problems with behaviors or feelings?".

Poor life chances validator

Life chances comprise factors related to quality of life (e.g., health, employability, education) (Tucker, 2016). Criminal convictions, substance use and school evasion were considered as indicators of this categorical composite validator.

Criminal conviction was considered positive to any of the following questions answered by parents/caregivers or the youth: "Has the youth ever received assistance from the assisted freedom program?", "Has the youth ever stayed overnight in a socio-educational center for adolescents?", or "Has the youth/ Have you ever been convicted of a crime?".

Substance use encompassed a range of self-reported behaviors including a high risk of alcohol misuse, habitual smoking, or the use of any illegal substances. The assessment for alcohol misuse risk utilized the Alcohol Use Disorders Identification Test Consumption (AUDIT-C), which has a scoring range from 0 to 12 (Bush et al., 1998). Participants with scores of 5 or above were categorized as being at a high risk for alcohol misuse. Tobacco dependence

was determined based on self-reports of daily smoking. The use of illegal substances was investigated through a confidential self-report questionnaire, where participants responded with a simple yes or no to a list of substances presented following the question, “In the last 12 months, have you used any of the following?”. The substances included cannabis, cocaine, crack, shoe glue, ecstasy (MDMA), LSD (or acid), solvents, and amphetamines.

School evasion was calculated based on school dropout or expulsion as reported by parents or caregivers since baseline.

Well-being

The evaluation of participants' well-being was conducted using the Short Warwick-Edinburgh Mental Well-being Scale (SWEMWBS) (Tennant et al., 2007), which comprises seven statements related to mental health and well-being and provides a continuous measure of this clinical validator. They are rated on a scale from 1 (none of the time) to 5 (all of the time), reflecting the past two weeks. The cumulative score derived from the sum of all item responses was used as a clinical validator in our analysis.

Covariates

We adjusted the analysis for the following possible confounding variables: age, sex (male as reference), race/ethnicity categorized as white or non-white (black, mixed and indigenous), study site (Porto Alegre as reference), social isolation and social class as classified by the Brazilian Association of Research Companies (ABEP). Social isolation was measured according to the item “I have at least one good friend” of the Strengths and Difficulties Questionnaire (SDQ). It was considered negative if the answer was “Not True” in a 3-item likert scale (0 = “Not True; 1 = Somewhat True; 2 = Certainly True) and otherwise true. The

ABEP is a parent-reported asset-based assessment that also includes parental education and sanitation to classify subjects as belonging to the A (highest) to E (lowest and reference category) socioeconomic class.

Oversampling and inverse probability weights

The BHRCS used an oversampling procedure, as described in Salum et. al, 2015. A sampling weight was generated to account for this high-risk stratification, which fully accounted for the oversampling procedure (Martel et al., 2017). In a previous analysis of the BHRCS, baseline maternal education and study site predicted response at follow-up (Ziebold et al., 2023). Furthermore, participants who had a mental health diagnosis at baseline and those with maternal mental health problems presented a higher proportion of missing data at follow-up (Tables S2 and S3). Therefore, we used these four variables to compute inverse probability weight (IPW) to address sample attrition in all regression analysis (Seaman and White, 2013). Sampling weight and IPW were multiplied, and weights for the few cases ($n = 36$) were trimmed for the 1st and 99th percentile. The resulting weights were used throughout the analyses to minimize bias associated with oversampling for high-risk procedures and missing data at follow-up, so the sample is closer to the general population at baseline.

Statistical analysis

Confirmatory factor analysis

Confirmatory factor analysis was used to evaluate if loneliness aggregates to other depressive symptoms in the SMFQ and to binarize the item “I felt lonely”, in order to distinguish a normative feeling from a clinical indicator, compatible with DAWBA-based depression items. The “I felt lonely” SMFQ item was dichotomized using a CFA threshold value of 0.5. The analysis was implemented with Weighted Least Square Mean and Variance

(WLSMV) as the estimation method. Model goodness of fit was evaluated using chi-square fit statistics, root-mean-square error of approximation (RMSEA), comparative fit index (CFI) and Tucker-Lewis index (TLI). A nonsignificant chi-square, RMSEA equal or below .06 and CFI and TLI above .95 indicate a good fit (Brown, 2015). In order to assess the reliability of the factors, we considered Lucke's omega (ω ; Lucke, 2005), a model-based reliability estimate, being analogous to alpha coefficient but appropriate for congeneric tests (varying factor loadings); Values of ω may vary between 0 and 1, where higher scores indicate greater reliability; a value of 1 indicates that the instrument's sum score measures the target construct with perfect accuracy.

Network Analysis

This study employed network analysis to evaluate symptom centrality according to SMFQ responses. Network models were estimated using regularized partial correlations, with graphical least absolute shrinkage and selection operator (GLASSO). Extended Bayesian Information criterion (EBIC) was used for model selection, with a γ tuning parameter of 0.5, to favor a more parsimonious model (Foygel and Drton, 2010). For this analysis, we used listwise deletion to deal with missing SFMQ data. Each item of the SMFQ was treated as a node in a network, and the relationships between these nodes were determined based on the pattern of responses. The strength of the connections between symptoms (nodes) was measured using partial correlation coefficients, controlling for the influence of other symptoms in the network. Expected influence, strength, betweenness, and closeness were computed to identify key symptoms that may play a significant role in depression. Our analysis was centered on expected influence, since it also accounts for the presence of negative edges (Robinaugh et al., 2016). Edge-weight accuracy was estimated with non-parametric bootstrapping ($n = 1000$), which was also used to test significant differences in edge-weight and strength (Epskamp and

Fried, 2018). Centrality stability was assessed using case-dropping bootstrapping procedures ($n = 1000$) to ensure the stability of the centrality estimates, which can be summarized with the correlation stability coefficient (CS-coefficient). The CS-coefficient measures the amount of data that can be dropped while still maintaining a correlation of 0.7 or greater with the original correlation coefficients, with 95% certainty. It should be ideally above 0.5 and at least above 0.25. To estimate the network and evaluate the model, we used the packages *bootnet* and *qgraph* for R (Version 4.3.1).

Regression models: depressive symptoms and clinical groups

We tested the independent association of the “I felt lonely” item, from the binarized SMFQ, with the clinical validators. In this model, the other nine depressive symptoms from DAWBA were included, using the DSM-like structure above-mentioned. We then modeled linear and logistic multiple regressions for the three clinical validators, adjusting for the covariates and weights. We tested multicollinearity by using the *vif* function from the “performance” package, which measures the variance inflation factor (VIF). We accepted a maximum VIF value of 10 (James et al., 2013; O’Brien, 2007).

The clinical groups were also used in multiple regression analysis to determine if the group with loneliness would be independently associated with the clinical validators, considering individuals depressed but not lonely as the reference group (direct comparison between participants with MDD with vs without loneliness). We regressed the life-threatening, poor life chances (and their components) and well-being validators on the groups. Analyses were adjusted for covariates and sampling weights.

Results

Loneliness as part of the depression construct and clinical operationalization

The unidimensional structure of SMFQ showed good fit indices (Table S5), with an ω reliability of 0.935, indicating strong unidimensionality. The item “I felt lonely” had a factor loading of 0.916, the second higher among tested depression symptoms. Given the established threshold of 0.5, loneliness was identified as a clinical indicator only when the corresponding item received a score of 2 (i.e., “True”). Therefore, those participants with score 0 or 1 were considered not lonely and those who scored 2 were considered positive for loneliness, as a clinical indicator for the regression analysis reported below.

Network analysis

Figure 1 represents the network structure based on SMFQ symptoms. The estimated network had 68 out of 78 possible non-zero edges. “I felt lonely” had a strong partial correlation with the items “I thought nobody really loved me”, “I cried a lot” and “I felt miserable or unhappy”. It was also the symptom with higher centrality, as measured by Expected Influence (Figure 2, Table S4). Variability of edge-weights can be seen in Figure S1. The bootstrap difference test between edge-weights showed that the relation between “I felt lonely” and “I felt miserable or unhappy” (a core depressive symptom) is significantly stronger than most other edge-weights (Figure S2). Besides that, “I felt lonely” is a significantly stronger node compared to almost all other nodes, except node 5 (“I felt I was no good anymore”) and node 8 (“I hated myself”) (Figure S3). Regarding centrality stability, Expected Influence had an optimal CS-coefficient ($CS(\text{cor} = 0.7) = 0.75$), which can be seen in Figure S4, demonstrating that regardless of sample size, the “I felt lonely” symptom keeps its centrality on the depressive symptom network.

[Figure 1 here]

[Figure 2 here]

Loneliness validity among other depression symptoms

Loneliness was independently associated with the life-threatening clinical validator (OR: 1.754; 95% CI: 1.187, 2.590). It was also associated with poor life chances (OR: 1.446; 95% CI: 1.065, 1.963) and negatively associated with well-being (B: -2.681; 95% CI: -3.792, -1.569). All results can be found on Table 1. Most items presented a multicollinearity index below 5. However, items representing the symptoms “Depressed/Irritable mood”, “Sleep changes”, “Tired/no energy” and “Poor concentration” had a VIF between 5 and 10 (Table S7), which is still acceptable (O’Brien, 2007).

[Table 1 here]

Loneliness as severity indicator - Sample characteristics and regressions by clinical groups

Sample characteristics by clinical groups are described in Table 2 (proportions were weighted for attrition and representativeness). We observed a higher prevalence of female participants across the clinical groups, particularly among those identified as depressed and lonely. Study site distribution varied, with São Paulo having a greater proportion of individuals solely with depression, whereas Porto Alegre had a higher count of participants experiencing both depression and loneliness. As expected, total SMFQ scores were higher in the clinical groups, especially among those feeling alone (groups 2 and 4).

[Table 2 here]

Table 3 demonstrates the association between life-threatening, poor life chances and well-being with three clinical groups compared to subjects diagnosed with depression which were not feeling lonely. Considering these clinical groups, those who were diagnosed with depression and were also lonely were at higher risks of having a life-threatening clinical validator (OR: 3.168; 95% CI: 2.039, 4.976) and scored lower on well-being (B: -3.474; 95% CI: -5.085, -1.862), while having similar odds of experiencing any poor life chances events when compared with those diagnosed with depression but were not feeling lonely (OR: 1.339; 95% CI: 0.873, 2.057). Analysis considering components of the life-threatening and poor life chances clinical indicators can be found in Tables S7 to S11. The conviction component of poor life chances validator was not evaluated separately, because of insufficient individuals for each clinical group.

[Table 3 here]

Discussion

This study aimed to replicate previous findings and expand the analysis on the concept that loneliness is a central depressive symptom in young people. Furthermore, it aimed to examine the potential to use loneliness as a severity indicator for MDD. Here, we showed that loneliness can be part of the latent depressive construct through CFA. We also replicated that loneliness is a central depressive symptom. Moreover, it is a symptom associated with poor life chances and life-threatening clinical validators as well as diminished well-being scores, above and beyond other diagnostic criteria. We also found that depressed individuals who also have loneliness as one of their symptoms present a higher association with recent life-threatening validators and present current lower well-being scores when compared with participants with MDD without loneliness.

Through CFA, we were able to operationalize loneliness as a clinical symptom and distinguish it from a normative feeling. Then, we could test loneliness as a clinical indicator to compare it with the other nine DSM symptoms of MDD. Compared to DSM symptoms, loneliness is the only item that was significantly associated with all three clinical validators, emphasizing its utility. Even though loneliness is not a DSM criteria for depression yet, those criteria present in the manual should not be confused “*with the disorders that they were designed to index*” (Kendler, 2016a). It is also important to evaluate symptoms not listed in the DSM (Kendler, 2016b) and loneliness appears to be useful to predict poor life outcomes in young depressed individuals (Viduani et al., 2024).

In line with previous findings (Gijzen et al., 2021; Manfro et al., 2021; Mullarkey et al., 2019), we were able to replicate that loneliness is an important symptom within the network of depressive symptoms, even when compared to other DSM criteria. Notably, the loneliness-sadness edge was one of the most central edges in the network, highlighting the relation between loneliness and a core depressive symptom according to the DSM-5-TR. The centrality of loneliness in a network of depressive symptoms also reflects the findings of Viduani *et. al.* (2024), where loneliness was the third most reported feature by young individuals with lived experience of depression.

Our findings suggest that loneliness is not only part of the syndrome and a useful symptom but may also serve as an indicator of severity in MDD. When viewed within the context of depressive symptomatology, individuals who experience loneliness and are diagnosed with MDD appear to be more vulnerable to life-threatening experiences and tend to have lower well-being scores compared to depressed individuals without this symptom. This is noteworthy because, unlike other medical conditions, mental health issues often lack clear biomarkers to evaluate disease severity, requiring clinical measures to assess it (Bentley et al., 2014; Kroenke et al., 2001; Zimmerman et al., 2018). In this regard, the DSM measures severity

based on the number and intensity of symptoms beyond the minimum required for diagnosis, without distinguishing between specific symptoms. Therefore, assessing loneliness in depressed individuals could aid clinicians in making risk assessment and intervention decisions.

Our study has several strengths, including the analysis of a large sample size. Additionally, by examining loneliness independently from social isolation, we controlled for potential confounding effects. Nonetheless, we acknowledge certain limitations. First, the employment of mixed evaluation tools (i.e., SMFQ and DAWBA) might limit the comprehensive evaluation of clinical operationalization of loneliness as a diagnostic criterion. This would need a new instrument that could test loneliness as a cardinal or conditional symptom in a structured interview such as DAWBA. Nonetheless, we used methods to generate a clinical operationalization of an unconditioned criteria and aggregate it with diagnostic criteria operationalized by a structured interview. Second, when assessing multicollinearity in the regression analysis by symptoms, some items showed VIF scores between 5 and 10, indicating moderate collinearity. Ideally, VIF should be below 5 and not exceed 10 (James et al., 2013; O'Brien, 2007). While still below our threshold, it is possible that higher VIF values are due to the dependency created between DAWBA items because of its skip-rule, a structural design choice of this questionnaire. Third, we relied on a cross-sectional design, which may not effectively predict future events but is useful for convergent validity. Life-threatening clinical validator assessed proximal (e.g., suicide attempt) and distant events (e.g., emergency visits) and poor life chances assessed lifetime events. However, our aim is not to attribute causal links but to test the association of these variables as they are associated with depression in bidirectional ways, as well as evaluate if both loneliness and other depressive symptoms are capturing clinically relevant information (e.g., people with past or current clinical risk). For instance, risk for suicide and depressive symptomatology remain high after a suicide attempt for months (Aaltonen et al., 2024) and several social determinants, such as convictions and

poor education are risk factors for MDD (Lund et al., 2018). Fourth, the study is limited in age range, which may not apply for middle-aged adults or elderly populations with depression. Fifth, this study is also limited to samples nested in a specific country and future studies should aim to replicate these findings in other contexts.

In conclusion, alongside previous literature, our research indicates that loneliness is central to depressive symptomatology in youth, is part of the construct and can be operationalized as a clinical criterion, which is informative for adverse life experiences, as well as it is an indicator of most severe forms of depression. Thus, loneliness assessment can be useful to inform depression and is a severity indicator above and beyond other symptoms. Future studies should aim to replicate these findings and test a formal operationalization of loneliness as a diagnostic criterion for MDD within the same structured interview, testing if it is a conditional or cardinal symptom and how many subjects are being left behind by not assessing loneliness as a MDD symptom.

Acknowledgements

Dr. João V. do Amaral receives a scholarship by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) – Finance Code 001

Gabriele dos Santos Jobim is supported by the FIPE/UFSM 2024 Júnior scholarship, granted to prof. Mauricio Scopel Hoffmann.

Rafaella Ormond Sampaio is supported by Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP 2022/16317-2).

Dr. Marcos Leite Santoro is supported by Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP 2023/05560-6).

Dr. Mauricio Scopel Hoffmann is supported by the United States National Institutes of Health grant R01MH120482 under his post-doctoral fellowship at UFRGS and by the Wellcome Mental Health Data Prize, granted by the Wellcome Trust (award reference 226697/Z/22/Z).

Funding sources

This work is supported by the National Institute of Developmental Psychiatry for Children and Adolescents (INPD), a science and technology institute funded by Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq; National Council for Scientific and Technological Development; grant numbers 573974/2008-0 and 465550/2014-2) and Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP; Research Support Foundation of the State of São Paulo; grant number 2008/57896-8, 2014/50917-0 and 2021/05332-8). This study was supported by the National Center for Research and Innovation in Mental Health (CISM), funded by FAPESP (grant number 2021/12901-9) and Banco Industrial do Brasil S/A. This work was also funded by the European Research Council under the European Union's Seventh Framework Programme (FP7/2007-2013)/ERC grant agreement no 337673.

Data Sharing

Data dictionary is available at <https://osf.io/ktz5h/wiki/Data%20Dictionaries/> and <https://osf.io/w3jr4> to direct download. Individual-level data is available upon request to the Brazilian High-Risk Cohort Study research committee, by following the instructions and filling the research form available at <https://osf.io/ktz5h/wiki/home/>. Study design and ethical details can be found elsewhere (Salum et al., 2015).

References

- Aaltonen, K., Sund, R., Hakulinen, C., Pirkola, S., Isometsä, E., 2024. Variations in Suicide Risk and Risk Factors After Hospitalization for Depression in Finland, 1996–2017. *JAMA Psychiatry* 81, 506. <https://doi.org/10.1001/jamapsychiatry.2023.5512>
- Abdellaoui, A., Nivard, M.G., Hottenga, J.-J., Fedko, I., Verweij, K.J.H., Baselmans, B.M.L., Ehli, E.A., Davies, G.E., Bartels, M., Boomsma, D.I., Cacioppo, J.T., 2018. Predicting loneliness with polygenic scores of social, psychological and psychiatric traits. *Genes Brain Behav.* 17, e12472.
- Abdellaoui, A., Verweij, K.J.H., 2021. Dissecting polygenic signals from genome-wide association studies on human behaviour. *Nat Hum Behav* 5, 686–694. <https://doi.org/10.1038/s41562-021-01110-y>
- Achterbergh, L., Pitman, A., Birken, M., Pearce, E., Sno, H., Johnson, S., 2020. The experience of loneliness among young people with depression: a qualitative meta-synthesis of the literature. *BMC Psychiatry* 20, 415.
- American Psychiatric Association (Ed.), 2022. Diagnostic and Statistical Manual of Mental Disorders: DSM-5-TR. American Psychiatric Association Publishing.
- Aristodemou, M.E., Kievit, R.A., Murray, A.L., Eisner, M., Ribeaud, D., Fried, E.I., 2024. Common Cause Versus Dynamic Mutualism: An Empirical Comparison of Two Theories of Psychopathology in Two Large Longitudinal Cohorts. *Clinical Psychological Science* 12, 380–402. <https://doi.org/10.1177/21677026231162814>
- Bentley, K.H., Gallagher, M.W., Carl, J.R., Barlow, D.H., 2014. Development and validation of the Overall Depression Severity and Impairment Scale. *Psychological Assessment* 26, 815–830. <https://doi.org/10.1037/a0036216>
- Beutel, M.E., Klein, E.M., Brähler, E., Reiner, I., Jünger, C., Michal, M., Wiltink, J., Wild, P.S., Münzel, T., Lackner, K.J., Tibubos, A.N., 2017. Loneliness in the general population: Prevalence, determinants and relations to mental health. *BMC Psychiatry* 17.
- Brown, T.A., 2015. Confirmatory factor analysis for applied research, 2nd ed., Confirmatory factor analysis for applied research, 2nd ed. The Guilford Press, New York, NY, US.
- Bryan, B.T., Thompson, K.N., Goldman-Mellor, S., Moffitt, T.E., Odgers, C.L., So, S.L.S., Uddin Rahman, M., Wertz, J., Matthews, T., Arseneault, L., 2024. The socioeconomic consequences of loneliness: Evidence from a nationally representative longitudinal study of young adults. *Soc. Sci. Med.* 116697.
- Bush, K., Kivlahan, D.R., McDonell, M.B., Fihn, S.D., Bradley, K.A., 1998. The AUDIT alcohol consumption questions (AUDIT-C): an effective brief screening test for problem drinking. Ambulatory Care Quality Improvement Project (ACQUIP). Alcohol Use Disorders Identification Test. *Arch. Intern. Med.* 158, 1789–1795.
- Cacioppo, J.T., Hughes, M.E., Waite, L.J., Hawkey, L.C., Thisted, R.A., 2006. Loneliness as a specific risk factor for depressive symptoms: cross-sectional and longitudinal analyses. *Psychol. Aging* 21, 140–151.
- Choi, K.W., Stein, M.B., Nishimi, K.M., Ge, T., Coleman, J.R.I., Chen, C.-Y., Ratanatharathorn, A., Zheutlin, A.B., Dunn, E.C., 23andMe Research Team, Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium, Breen, G., Koenen, K.C., Smoller, J.W., 2020. An Exposure-Wide and Mendelian Randomization

- Approach to Identifying Modifiable Factors for the Prevention of Depression. *Am. J. Psychiatry* 177, 944–954.
- Epskamp, S., Fried, E.I., 2018. A tutorial on regularized partial correlation networks. *Psychol. Methods* 23, 617–634.
- Erzen, E., Çikrikci, Ö., 2018. The effect of loneliness on depression: A meta-analysis. *International Journal of Social Psychiatry*.
- Fleitlich-Bilyk, B., Goodman, R., 2004. Prevalence of Child and Adolescent Psychiatric Disorders in Southeast Brazil. *Journal of the American Academy of Child & Adolescent Psychiatry* 43, 727–734. <https://doi.org/10.1097/01.chi.0000120021.14101.ca>
- Foygel, R., Drton, M., 2010. Extended Bayesian Information Criteria for Gaussian Graphical Models.
- Fusar-Poli, P., Estradé, A., Stanghellini, G., Esposito, C.M., Rosfort, R., Mancini, M., Norman, P., Cullen, J., Adesina, M., Jimenez, G.B., da Cunha Lewin, C., Drah, E.A., Julien, M., Lamba, M., Mutura, E.M., Prawira, B., Sugianto, A., Teresa, J., White, L.A., Damiani, S., Vasconcelos, C., Bonoldi, I., Politi, P., Vieta, E., Radden, J., Fuchs, T., Ratcliffe, M., Maj, M., 2023. The lived experience of depression: a bottom-up review co-written by experts by experience and academics. *World Psychiatry* 22, 352–365.
- Gariépy, G., Honkaniemi, H., Quesnel-Vallée, A., 2016. Social support and protection from depression: systematic review of current findings in Western countries. *Br. J. Psychiatry* 209, 284–293.
- Gijzen, M.W.M., Rasing, S.P.A., Creemers, D.H.M., Smit, F., Engels, R.C.M.E., De Beurs, D., 2021. Suicide ideation as a symptom of adolescent depression. a network analysis. *J. Affect. Disord.* 278, 68–77.
- Goodman, R., Ford, T., Richards, H., Gatward, R., Meltzer, H., 2000. The Development and Well-Being Assessment: Description and initial validation of an integrated assessment of child and adolescent psychopathology. *J. Child Psychol. Psychiatry* 41, 645–655.
- Hodgson, S., Watts, I., Fraser, S., Roderick, P., Dambha-Miller, H., 2020. Loneliness, social isolation, cardiovascular disease and mortality: a synthesis of the literature and conceptual framework. *J. R. Soc. Med.* 113, 185–192.
- Holt-Lunstad, J., 2022. Social connection as a public health issue: The evidence and a systemic framework for prioritizing the “social” in social determinants of health. *Annu. Rev. Public Health* 43, 193–213.
- Holt-Lunstad, J., Perissinotto, C., 2023. Social Isolation and Loneliness as Medical Issues. *N. Engl. J. Med.* 388, 193–195.
- Holt-Lunstad, J., Smith, T.B., Baker, M., Harris, T., Stephenson, D., 2015. Loneliness and Social Isolation as Risk Factors for Mortality: A Meta-Analytic Review. *Perspect. Psychol. Sci.* 10, 227–237.
- ICD-11 for Mortality and Morbidity Statistics [WWW Document]. URL <https://icd.who.int/browse/2024-01/mms/en> (accessed 11.11.24).
- James, G., Witten, D., Hastie, T., Tibshirani, R., 2013. *An Introduction to Statistical Learning*. Springer US.
- Kendler, K.S., 2016a. Phenomenology of Schizophrenia and the Representativeness of Modern Diagnostic Criteria. *JAMA Psychiatry* 73, 1082–1092.

- Kendler, K.S., 2016b. The Phenomenology of Major Depression and the Representativeness and Nature of DSM Criteria. *Am. J. Psychiatry* 173, 771–780.
- Kroenke, K., Spitzer, R.L., Williams, J.B., 2001. The PHQ-9: validity of a brief depression severity measure. *J. Gen. Intern. Med.* 16, 606–613.
- Leigh-Hunt, N., Bagguley, D., Bash, K., Turner, V., Turnbull, S., Valtorta, N., Caan, W., 2017. An overview of systematic reviews on the public health consequences of social isolation and loneliness. *Public Health* 152, 157–171.
- Lim, M.H., Eres, R., Vasan, S., 2020. Understanding loneliness in the twenty-first century: an update on correlates, risk factors, and potential solutions. *Soc. Psychiatry Psychiatr. Epidemiol.* 55, 793–810.
- Lucke, J.F., 2005. The α and the ω of Congeneric Test Theory: An Extension of Reliability and Internal Consistency to Heterogeneous Tests. *Applied Psychological Measurement* 29, 65–81. <https://doi.org/10.1177/0146621604270882>
- Luhmann, M., Buecker, S., Rüsberg, M., 2023. Loneliness across time and space. *Nat Rev Psychol* 2, 9–23.
- Lund, C., Brooke-Sumner, C., Baingana, F., Baron, E.C., Breuer, E., Chandra, P., Haushofer, J., Herrman, H., Jordans, M., Kieling, C., Medina-Mora, M.E., Morgan, E., Omigbodun, O., Tol, W., Patel, V., Saxena, S., 2018. Social determinants of mental disorders and the Sustainable Development Goals: a systematic review of reviews. *Lancet Psychiatry* 5, 357–369.
- Luo, Y., Hawkey, L.C., Waite, L.J., Cacioppo, J.T., 2012. Loneliness, health, and mortality in old age: A national longitudinal study. *Social Science and Medicine* 74, 907–914.
- Manfro, P.H., Pereira, R.B., Rosa, M., Cogo-Moreira, H., Fisher, H.L., Kohrt, B.A., Mondelli, V., Kieling, C., 2021. Adolescent depression beyond DSM definition: a network analysis. *Eur. Child Adolesc. Psychiatry*.
- Mann, F., Wang, J., Pearce, E., Ma, R., Schlieff, M., Lloyd-Evans, B., Ikhtabi, S., Johnson, S., 2022. Loneliness and the onset of new mental health problems in the general population. *Soc. Psychiatry Psychiatr. Epidemiol.* 57, 2161–2178.
- Martel, M.M., Pan, P.M., Hoffmann, M.S., Gadelha, A., do Rosário, M.C., Mari, J.J., Manfro, G.G., Miguel, E.C., Paus, T., Bressan, R.A., Rohde, L.A., Salum, G.A., 2017. A general psychopathology factor (P factor) in children: Structural model analysis and external validation through familial risk and child global executive function. *J. Abnorm. Psychol.* 126, 137–148.
- Midgley, N., Parkinson, S., Holmes, J., Stapley, E., Eatough, V., Target, M., 2015. Beyond a diagnosis: The experience of depression among clinically-referred adolescents. *J. Adolesc.* 44, 269–279.
- Mullarkey, M.C., Marchetti, I., Beevers, C.G., 2019. Using Network Analysis to Identify Central Symptoms of Adolescent Depression. *J. Clin. Child Adolesc. Psychol.* 48, 656–668.
- O’Brien, R.M., 2007. A Caution Regarding Rules of Thumb for Variance Inflation Factors. *Qual. Quant.* 41, 673–690.
- Perlman, D., Peplau, L.A., 1981. Toward a social psychology of loneliness. *Pers. Relatsh.* 3, 31–56.
- Pinto, E.R.S. [unifesp], 2014. Short Mood and Feelings Questionnaire: tradução para língua portuguesa, adaptação cultural e validação.

- Robinaugh, D.J., Millner, A.J., McNally, R.J., 2016. Identifying highly influential nodes in the complicated grief network. *J. Abnorm. Psychol.* 125, 747–757.
- Salum, G.A., Gadelha, A., Pan, P.M., Moriyama, T.S., Graeff-Martins, A.S., Tamanaha, A.C., Alvarenga, P., Valle Krieger, F., Fleitlich-Bilyk, B., Jackowski, A., Sato, J.R., Brietzke, E., Polanczyk, G.V., Brentani, H., de Jesus Mari, J., Do Rosário, M.C., Manfro, G.G., Bressan, R.A., Mercadante, M.T., Miguel, E.C., Rohde, L.A., 2015. High risk cohort study for psychiatric disorders in childhood: rationale, design, methods and preliminary results. *Int. J. Methods Psychiatr. Res.* 24, 58–73.
- Sbarra, D.A., Ramadan, F.A., Choi, K.W., Treur, J.L., Levey, D.F., Wootton, R.E., Stein, M.B., Gelernter, J., Klimentidis, Y.C., 2023. Loneliness and depression: bidirectional mendelian randomization analyses using data from three large genome-wide association studies. *Mol. Psychiatry*.
- Seaman, S.R., White, I.R., 2013. Review of inverse probability weighting for dealing with missing data. *Stat. Methods Med. Res.* 22, 278–295.
- Solmi, M., Veronese, N., Galvano, D., Favaro, A., Ostinelli, E.G., Noventa, V., Favaretto, E., Tudor, F., Finessi, M., Shin, J.I., Smith, L., Koyanagi, A., Cester, A., Bolzetta, F., Cotroneo, A., Maggi, S., Demurtas, J., De Leo, D., Trabucchi, M., 2020. Factors Associated With Loneliness: An Umbrella Review Of Observational Studies. *J. Affect. Disord.* 271, 131–138.
- Surkalim, D.L., Luo, M., Eres, R., Gebel, K., van Buskirk, J., Bauman, A., Ding, D., 2022. The prevalence of loneliness across 113 countries: systematic review and meta-analysis. *BMJ* 376, e067068.
- Tennant, R., Hiller, L., Fishwick, R., Platt, S., Joseph, S., Weich, S., Parkinson, J., Secker, J., Stewart-Brown, S., 2007. The Warwick-Edinburgh Mental Well-being Scale (WEMWBS): development and UK validation. *Health Qual. Life Outcomes* 5, 63.
- Thapar, A., Eyre, O., Patel, V., Brent, D., 2022. Depression in young people. *Lancet* 400, 617–631.
- Tucker, J., 2016. Improving Children's life chances. Child Poverty Action Group.
- Valtorta, N.K., Kanaan, M., Gilbody, S., Ronzi, S., Hanratty, B., 2016. Loneliness and social isolation as risk factors for coronary heart disease and stroke: systematic review and meta-analysis of longitudinal observational studies. *Heart* 102, 1009–1016.
- Viduani, A., Arenas, D.L., Benetti, S., Wahid, S.S., Kohrt, B.A., Kieling, C., 2024. Systematic Review and Meta-Synthesis: How Is Depression Experienced by Adolescents? A Synthesis of the Qualitative Literature. *J. Am. Acad. Child Adolesc. Psychiatry*.
- Viduani, A., Benetti, S., Martini, T., Buchweitz, C., Ottman, K., Wahid, S.S., Fisher, H.L., Mondelli, V., Kohrt, B.A., Kieling, C., 2021. Social isolation as a core feature of adolescent depression: a qualitative study in Porto Alegre, Brazil. *Int. J. Qual. Stud. Health Well-being*. 16.
- Wang, F., Gao, Y., Han, Z., Yu, Y., Long, Z., Jiang, X., Wu, Y., Pei, B., Cao, Y., Ye, J., Wang, M., Zhao, Y., 2023. A systematic review and meta-analysis of 90 cohort studies of social isolation, loneliness and mortality. *Nat Hum Behav*.
- Wang, J., Mann, F., Lloyd-Evans, B., Ma, R., Johnson, S., 2018. Associations between loneliness and perceived social support and outcomes of mental health problems: A systematic review. *BMC Psychiatry* 18.

- Zachar, P., Kendler, K.S., 2017. The Philosophy of Nosology. *Annu. Rev. Clin. Psychol.* 13, 49–71.
- Ziebold, C., Evans-Lacko, S., Andrade, M.C.R., Hoffmann, M., Fonseca, L., Barbosa, M., Pan, P.M., Miguel, E., Bressan, R., Rohde, L.A., Salum, G., Schafer, J., de Jesus Mari, J., Gadelha, A., 2023. Childhood poverty and mental health disorders in early adulthood: evidence from a Brazilian cohort study. *Eur. Child Adolesc. Psychiatry* 32, 903–914.
- Zimmerman, M., Balling, C., Chelminski, I., Dalrymple, K., 2018. Understanding the severity of depression: Which symptoms of depression are the best indicators of depression severity? *Compr. Psychiatry* 87, 84–88.

Tables and figures

Table 1 - Multiple regression models by depressive symptoms for each clinical validator

Variable	Any life-threatening validator			Any life chances validator			Well-being		
	OR	95% CI	p	OR	95% CI	p	B	95% CI	p
<i>Depressive Symptoms</i>									
Depressed/Irritable mood	0.735	0.383, 1.395	0.348	0.754	0.472, 1.199	0.234	0.849	-0.806, 2.504	0.314
Loss of interest	1.480	0.888, 2.475	0.133	0.891	0.598, 1.326	0.569	-0.223	-1.672, 1.227	0.763
Weight loss/gain or changed appetite	1.331	0.776, 2.285	0.298	4.187	2.749, 6.458	<0.001	-1.615	-3.097, -0.133	0.033
Sleep changes	0.517	0.232, 1.157	0.108	1.715	1.003, 2.986	0.052	-1.523	-3.388, 0.342	0.109
Agitation	1.809	1.172, 2.806	0.008	1.187	0.829, 1.701	0.349	-0.220	-1.528, 1.088	0.742
Tired/no energy	1.314	0.619, 2.812	0.478	0.303	0.177, 0.509	<0.001	-1.346	-3.160, 0.469	0.146
Feels worthless, guilty	0.895	0.516, 1.560	0.694	0.952	0.629, 1.443	0.817	-0.661	-2.162, 0.840	0.388
Poor concentration	0.628	0.324, 1.207	0.164	0.992	0.630, 1.564	0.972	0.377	-1.255, 2.008	0.651
Suicidal Ideation	34.122	23.069, 51.230	<0.001	1.128	0.848, 1.497	0.406	-1.530	-2.559, -0.502	0.004
Loneliness	1.754	1.187, 2.590	0.005	1.446	1.065, 1.963	0.018	-2.681	-3.792, -1.569	<0.001
<i>Covariates</i>									
Age (years) at domicile evaluation	1.169	1.084, 1.262	<0.001	1.384	1.328, 1.444	<0.001	0.076	-0.070, 0.222	0.307
Socioeconomic status (class C vs D/E)	0.749	0.501, 1.128	0.162	0.521	0.414, 0.656	<0.001	0.881	0.018, 1.744	0.045

Socioeconomic status (class A/B vs D/E)	1.482	0.933, 2.372	0.098	0.412	0.315, 0.538	<0.001	1.695	0.710, 2.679	<0.001
Gender: Male	0.619	0.448, 0.854	0.004	2.176	1.835, 2.584	<0.001	0.520	-0.079, 1.118	0.089
Study site: São Paulo	2.101	1.545, 2.877	<0.001	0.584	0.497, 0.686	<0.001	-0.403	-0.984, 0.179	0.174
Race/ethnicity: White	0.541	0.397, 0.735	<0.001	0.796	0.676, 0.938	0.007	-0.399	-0.992, 0.195	0.188
Social isolation	1.046	0.672, 1.605	0.841	1.110	0.859, 1.431	0.423	-1.667	-2.598, -0.735	<0.001
Observations	1,188			1,188			1,188		
Pseudo R ² / R ²	0.379			0.168			0.226		

Note: OR, odds ratio; CI, confidence interval; Bold, significant associations of depressive symptoms.

Table 2 - Sample characteristics by clinical groups

Characteristic	Neither depressed nor lonely, N = 923	Not depressed but lonely, N = 68	Depressed but not lonely, N = 105	Depressed and Lonely, N = 92	p-value
Age in years	17.82 (1.99)	17.65 (1.95)	18.41 (2.04)	17.96 (1.84)	0.2
Gender					<0.001
Female	380 (42%)	42 (59%)	62 (62%)	66 (72%)	
Male	543 (58%)	26 (41%)	43 (38%)	26 (28%)	
Race/ethnicity					0.200
Non-White	377 (41%)	26 (33%)	31 (29%)	33 (38%)	
White	546 (59%)	42 (67%)	74 (71%)	59 (62%)	
Study site					0.024
RS	441 (47%)	33 (58%)	61 (64%)	42 (39%)	
SP	482 (53%)	35 (42%)	44 (36%)	50 (61%)	
Socioeconomic status					0,003
Class A/B	122 (12%)	7 (8.1%)	13 (15%)	23 (33%)	
Class C	584 (63%)	45 (69%)	60 (55%)	50 (46%)	
Class D/E	217 (25%)	16 (23%)	32 (30%)	19 (21%)	
Total SMFQ Score	3.0 (3.5)	15.5 (4.7)	7.9 (5.1)	19.0 (4.3)	<0.001
Isolated	109 (11%)	9 (14%)	22 (11%)	16 (9.6%)	0,900
Any life-threatening validator	57 (6.0%)	16 (22%)	27 (23%)	53 (51%)	<0.001
Self-harm	35 (3.7%)	14 (19%)	22 (19%)	48 (49%)	<0.001
Admission in psychiatric facility	7 (0.8%)	1 (2.3%)	4 (4.7%)	2 (1.9%)	0.066
ER visit due to mental health issues	20 (2.4%)	4 (5.0%)	6 (4.7%)	10 (7.8%)	0.052
Any poor life chances validator	360 (38%)	25 (39%)	54 (51%)	53 (53%)	0.067
Conviction	36 (3.8%)	0 (0%)	5 (3.5%)	2 (<0.1%)	0.200
Substance use in past	253 (28%)	18 (30%)	42 (40%)	45 (44%)	0.041

year					
School evasion	183 (18%)	16 (20%)	35 (28%)	28 (27%)	0.130
Well Being Score	25.8 (5.2)	21.5 (5.0)	21.6 (4.9)	17.8 (5.1)	<0.001

Note: Mean (SD); n (unweighted) (% weighted); Wilcoxon rank-sum test for complex survey samples; chi-squared test with Rao & Scott's second-order correction

Table 3 - Regression analysis: clinical groups associated with clinical validators

Variable	Any life-threatening validator			Any poor life-chances validator			Well-being		
	OR	95% CI	p	OR	95% CI	p	β	95% CI	p
Predictors									
Neither depressed nor lonely	0.214	0.147, 0.315	<0.001	0.620	0.452, 0.849	0.003	4.060	2.873, 5.248	<0.001
Not depressed but lonely	1.013	0.597, 1.703	0.963	0.786	0.495, 1.241	0.303	-0.043	-1.780, 1.694	0.961
Depressed and lonely	3.168	2.039, 4.976	<0.001	1.339	0.873, 2.057	0.181	-3.474	-5.085, -1.862	<0.001
Age (years) at domicile evaluation	1.139	1.067, 1.216	<0.001	1.378	1.322, 1.436	<0.001	0.067	-0.082, 0.215	0.379
Socioeconomic status (class C vs D/E)	0.752	0.539, 1.059	0.098	0.522	0.415, 0.656	<0.001	0.897	0.017, 1.777	0.046
Socioeconomic status (class A/B vs D/E)	1.335	0.907, 1.975	0.145	0.417	0.320, 0.544	<0.001	1.671	0.668, 2.674	0.001
Gender: Male	0.475	0.363, 0.619	<0.001	2.031	1.726, 2.395	<0.001	0.886	0.291, 1.481	0.004
Study site: São Paulo	1.691	1.302, 2.204	<0.001	0.583	0.497, 0.684	<0.001	-0.352	-0.943, 0.240	0.244
Race/ethnicity: White	0.604	0.465, 0.785	<0.001	0.840	0.715, 0.987	0.034	-0.428	-1.030, 0.174	0.163
Social isolation	1.760	1.224, 2.493	0.002	1.117	0.869, 1.432	0.386	-1.911	-2.849, -0.973	<0.001
Observations	1,188			1,188			1,188		
Pseudo R ² / R ²	0.226			0.156			0.194		

Note: OR, odds ratio; CI, confidence interval;

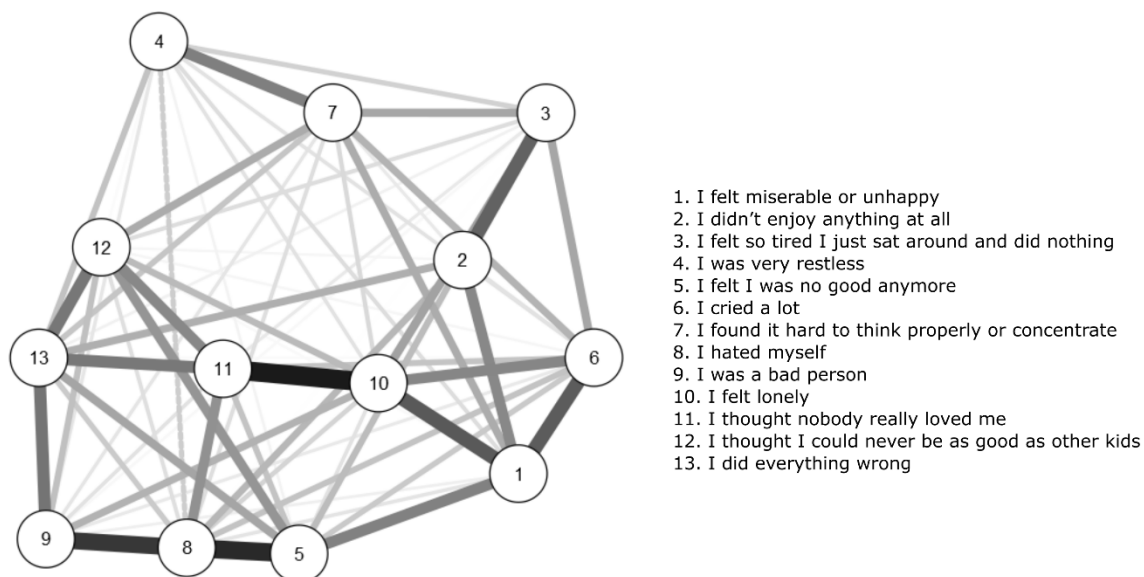


Figure 1. Estimated network structure of SMFQ symptoms.

[network_figure_1]

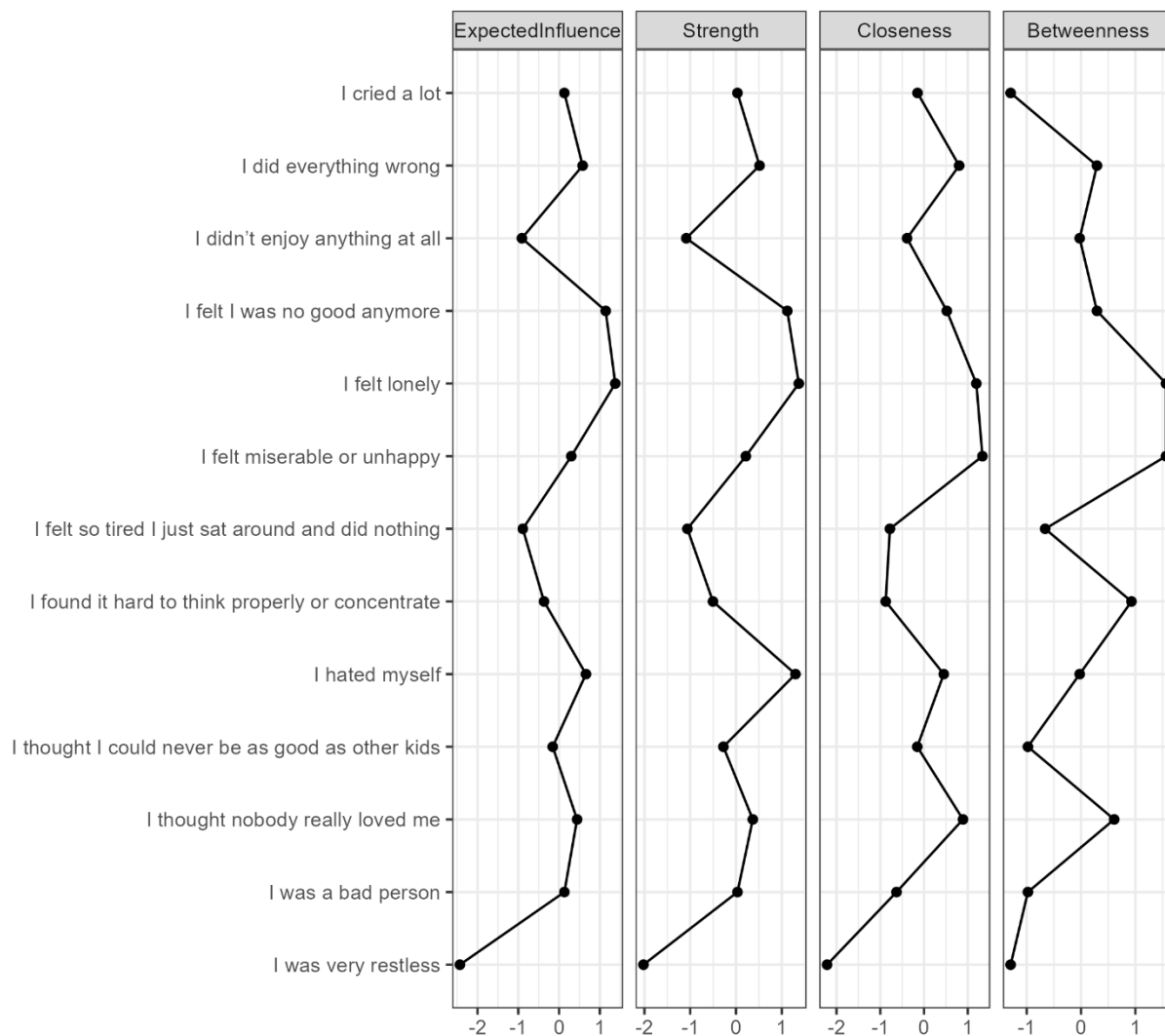


Figure 2. Centrality indices of SMQF items, shown as standardized z-scores.

[centrality_ind_figure_2]